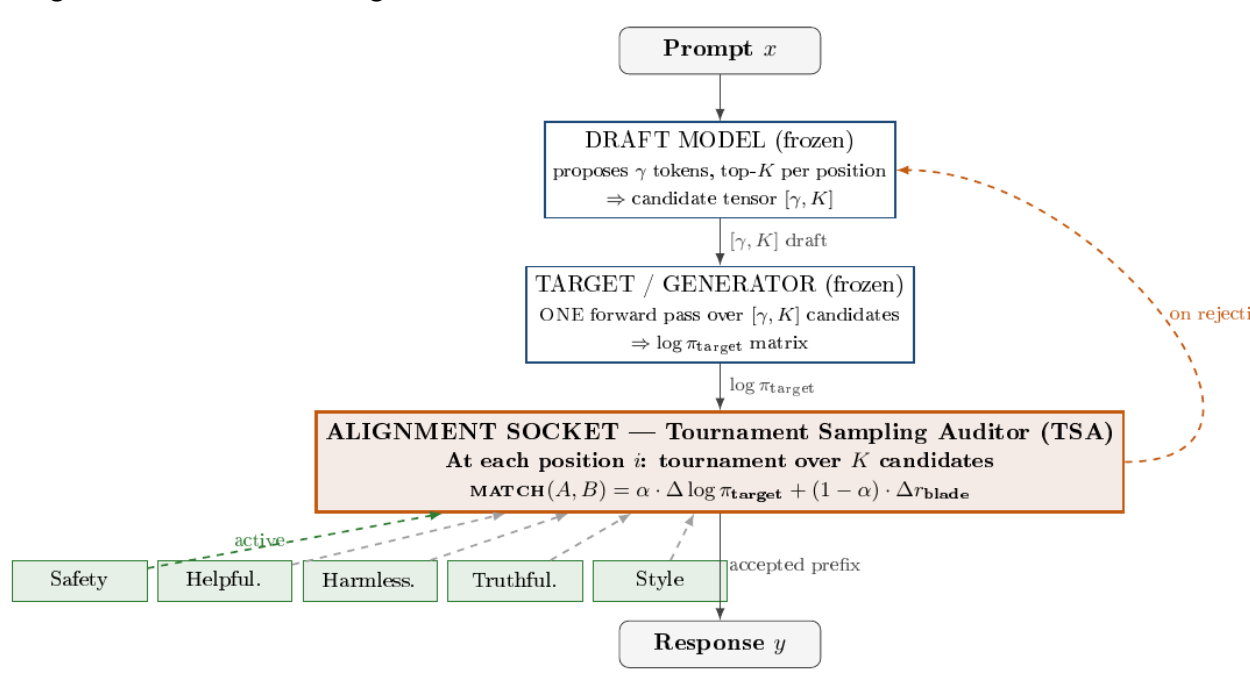


Ideas

The problem with deterministic tournament scoring

In the original Swiss Knife proposal, the alignment head (the "blade") produces a deterministic scalar reward for each candidate token — call it r_{blade} . The tournament was designed so that every candidate plays every round and the winner emerges through pairwise comparisons rather than a single argmax. The problem: if every match uses the same fixed scores, the pairwise comparisons add no information. A candidate that has the highest score beats every opponent it faces, in every round, regardless of the bracket structure. The tournament collapses to argmax selection. The elaborate Swiss pairing logic becomes irrelevant.

Original architecture diagram:



For the tournament to do something the argmax cannot, the scoring function must behave differently across matches. Two properties are needed:

- Stochasticity — the score of a candidate is a random variable, not a fixed scalar. A lower-quality candidate can occasionally win a match.
- Intransitivity — because each match draws an independent random functional, the win relation is not transitive. Candidate A can beat B, B can

beat C, and C can beat A. This is only meaningful in a pairwise tournament where the bracket structure matters.

Both properties vanish the moment the same score vector is used for every match.

The fix:

Instead of scoring candidates with a single fixed forward pass of the blade, we define a family of stochastic scalar functionals of the blade's internal state. Before each individual match in the tournament, a fresh member of this family is sampled and a new random perturbation is applied to the blade's computation. After the match, the perturbation is removed.

1. MC Dropout

Before the final linear projection, a fresh binary dropout mask is applied to the hidden state vector h :

$$h_{\text{match}} = \text{Dropout}(h, p)$$

Neurons zeroed by the mask contribute nothing to the logit distribution for this match. The specific neurons zeroed are resampled independently for every match.

What this tests: whether a candidate's reward advantage is robust to noise in the final feature readout, or whether it is carried by a small set of co-firing hidden neurons that may not consistently dominate in every match.

2. Random Projection

A random matrix R (normalized) is drawn per match and used to perturb the hidden state before projection:

$$h_{\text{match}} = h + \varepsilon \cdot (hR)$$

This rotates and scales the representation in a fresh random direction for every match, keeping the magnitude controlled via ε .

What this tests: whether the reward signal is stable under local linear perturbations of the representation geometry.

3. Attention Head Subsampling

In each match, a random subset of attention heads in the final N transformer layers are silenced (their outputs are zeroed). The specific subset is redrawn per match.

Different attention heads specialise in different kinds of context (e.g., syntactic coherence, coreference, safety-relevant patterns).

By randomly varying which heads contribute to the reward signal, each match is effectively judged by a different committee of "attentional criteria." A candidate that is preferred regardless of which subset of heads is active has a more broadly grounded alignment signal.

How these integrate into the Swiss Knife proposal

The Swiss-system tournament runs $R = \lceil \log_2 K \rceil$ rounds. In each round, candidates are paired by their current cumulative win count (Swiss pairing rule).

For each pair (A, B) , the auditor:

1. Draws a fresh functional from the chosen family.
2. Scores all K candidates under that perturbed forward pass.
3. Determines the winner of the match using $\alpha \cdot \Delta_{\text{target}} + (1 - \alpha) \cdot \Delta_{\text{blade}}$.

Because the functional is redrawn independently for every match, the score $r_{\text{blade}}(k)$ for candidate k is a different random variable in each match. This directly produces the intransitivity and stochasticity needed to justify the tournament structure.

